

LA HW

Included exercises: 1, 3, 5, 7–22, 55, 57, 59, 61, 63, 65, 67, 69, 71, 73, 75

Exercises 1–6: Coefficient and Augmented Matrices

For each system, write (a) its coefficient matrix and (b) its augmented matrix.

Original Exercise 1

$$\begin{aligned} -x_2 + 2x_3 &= 0, \\ x_1 + 3x_2 &= -1. \end{aligned}$$

Original Exercise 3

$$\begin{aligned} x_1 + 2x_2 &= 3, \\ -x_1 + 3x_2 &= 2, \\ -3x_1 + 4x_2 &= 1. \end{aligned}$$

Original Exercise 5

$$\begin{aligned} 2x_2 - 3x_3 &= 4, \\ -x_1 + x_2 + 2x_3 &= -6, \\ 2x_1 + x_3 &= 0. \end{aligned}$$

Exercises 7–14: Elementary Row Operations

Perform the indicated elementary row operation on

$$\begin{bmatrix} 1 & -1 & 0 & 2 & -3 \\ -2 & 6 & 3 & -1 & 1 \\ 0 & 2 & -4 & 4 & 2 \end{bmatrix}.$$

Original Exercise 7 Interchange rows 1 and 3.**Original Exercise 8** Multiply row 1 by -3 .**Original Exercise 9** Add 2 times row 1 to row 2.**Original Exercise 10** Interchange rows 1 and 2.**Original Exercise 11** Multiply row 3 by $\frac{1}{2}$.**Original Exercise 12** Add -3 times row 3 to row 2.

Original Exercise 13 Add 4 times row 2 to row 3.

Original Exercise 14 Add 2 times row 1 to row 3.

Exercises 15–22: Elementary Row Operations

Perform the indicated elementary row operation on

$$\begin{bmatrix} 1 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & -4 & 6 \\ -3 & 2 & 1 \end{bmatrix}.$$

Original Exercise 15 Multiply row 1 by -2 .

Original Exercise 16 Multiply row 2 by $\frac{1}{2}$.

Original Exercise 17 Add -2 times row 1 to row 3.

Original Exercise 18 Add 3 times row 1 to row 4.

Original Exercise 19 Interchange rows 2 and 3.

Original Exercise 20 Interchange rows 2 and 4.

Original Exercise 21 Add -2 times row 2 to row 4.

Original Exercise 22 Add 2 times row 2 to row 1.

Exercise 55: Basic and Free Variables

Answer the question and explain your reasoning.

Original Exercise 55 Suppose that the general solution of a system of m linear equations in n variables contains k free variables. How many basic variables does it have? Explain your answer.

Exercises 57–76: True or False

Determine whether each statement is true or false.

Original Exercise 57 Every system of linear equations has at least one solution.

Original Exercise 59 If a matrix A can be transformed into a matrix B by an elementary row operation, then B can be transformed into A by an elementary row operation.

Original Exercise 61 If a matrix is in reduced row echelon form, then the leading entry of each nonzero row is 1.

Original Exercise 63 Every matrix can be transformed into a unique matrix in row echelon form by a sequence of elementary row operations.

Original Exercise 65 Performing an elementary row operation on the augmented matrix of a system of linear equations produces the augmented matrix of an equivalent system of linear equations.

Original Exercise 67 If the only nonzero entry in some row of an augmented matrix of a system of linear equations lies in the last column, then the system is inconsistent.

Original Exercise 69 If A is the coefficient matrix of a system of m linear equations in n variables, then A is an $n \times m$ matrix.

Original Exercise 71 If the reduced row echelon form of the augmented matrix of a consistent system of m linear equations in n variables contains k nonzero rows, then its general solution contains k basic variables.

Original Exercise 73 Multiplying every entry of some row of a matrix by a scalar is an elementary row operation.

Original Exercise 75 If a system of linear equations has more variables than equations, then it must have infinitely many solutions.